Phase 5 Contribution and AI sheet.
Phase 5 due 3/15/2026

The total number of files is 10 + updated assembler, so 11 total .v files.

Pipeline Registers

- `IF_ID.v` — holds PC and instruction between Fetch and Decode
- `ID_EX.v` — holds control signals, register values, immediates between Decode and Execute
- `EX_MEM.v` — holds ALU result, write data, control signals between Execute and Memory
- `MEM_WB.v` — holds memory/ALU result and control signals between Memory and Writeback

Hazard Handling

- `FORWARDING_UNIT.v` — detects RAW hazards and generates ForwardA/ForwardB select signals
- `HAZARD_DETECTION.v` — detects load-to-use hazards and generates stall/flush signals

New Muxes

- `MUX_3_1.v` — you'll need 3-to-1 muxes for the forwarding paths into the ALU (ForwardA and ForwardB selection)

Existing files you'll likely need to significantly modify:

- `RISCV_TOP.v` — rewire everything to connect pipeline stages
- `CONTROL.v` — control signals now need to flow through pipeline registers
- `REGISTER.v` — may need adjustments for stalling

That's about 7 new files. Want to start with the pipeline registers since everything else depends on them?

Amanda - MUX_3_1.v file, IF_ID.v file, Contribution/AI document

Sophia - REGISTER.v file, HAZARD_DETECTION.v file, FORWARDING_UNIT.v file

Evan - CONTROL.v file, ID_EX.v file, Testing and submission

Banmeet - RISCV_TOP.v file, MEM_WB.v file, EX_MEM.v file